ST MARY'S DSG, KLOOF

GRADE: 10 DATE: 30 AUGUST 2023

PHYSICAL SCIENCE OPEN BOOK TEST
KINEMATICS

TIME: 43 MINS TOTAL: 51

ASSESSOR: K. Bolton MODERATORS: G. Main and T. Kirsten

NAME:_____ **TEACHER:**_____

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 6. Please check that your question paper is complete. A data sheet is provided.

2. Answer the questions on your question paper.

3. Work to 2 decimal places for final answers.

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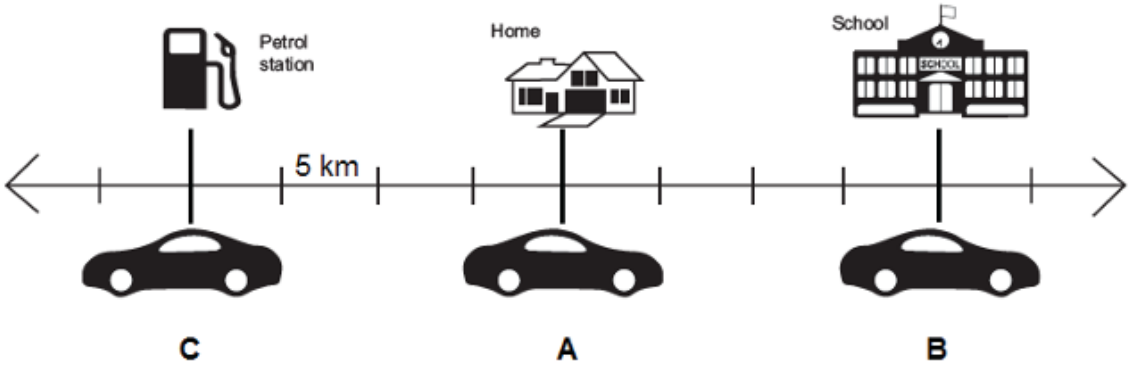
1. Peter walks to school 400 m **eastwards** down the street, and then he turns north and walks another 300 m due **north**.
   a. Draw a head to tail diagram to illustrate Peter's resultant displacement. (2)

   b. Calculate the magnitude of Peter's resultant displacement. (2)

c. How far does he walk to school? (2)

d. When he returns home in the afternoon, what is his resultant displacement? (2)

2. Ted drives his car along the road, starting at point **A** and driving to point **B**, in 30 minutes. He then stops and drives from point **B** to point **C** in 45 minutes. Take the positive direction as moving to the **right.**
Scale: Each interval represents 5 km.



Calculate, in $km \cdot h^{-1}$:

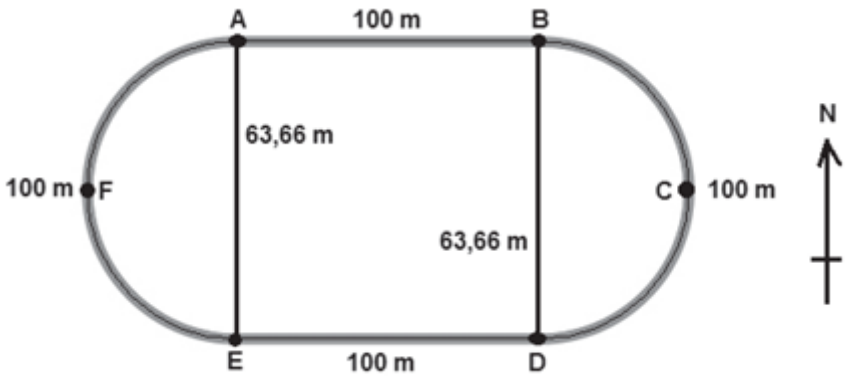
a. Ted's average speed from A to B. (3)

b. Ted's average velocity from A to B. (2)

c. Ted's average speed from B to C. (3)

d. Ted's average velocity from B to C. (3)

3. An athlete runs **clockwise** from A to B to C to D to E on a 400 m oval athletic track at a constant speed of 5 $m \cdot s^{-1}$.



a. How long does she take to run from A to E along the track? (2)

b. Describe her displacement relative to point A when she reaches E. (2)

c. Calculate the magnitude of her average velocity when she reaches point E. (2)

d. Determine her instantaneous velocity at point C. (2)

4. A tractor, travelling **east**, accelerates uniformly from rest to 4 $m \cdot s^{-1}$ in 20 s. It then travels at a steady speed for 440 m, and finally comes to rest 10 s later, having travelled a total of 500 m.

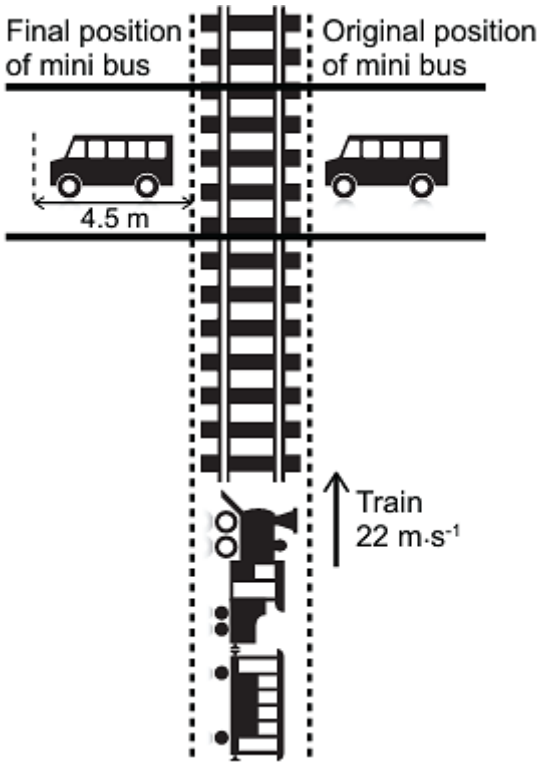
a. Calculate the acceleration of the tractor during the first 20 s. (3)

b. Calculate the magnitude of the tractor's displacement in the first 20 s. (3)

c. Calculate the magnitude of the tractor's average velocity for the entire period of its motion. (3)

d. Calculate the acceleration of the tractor during the last 10 s. (3)

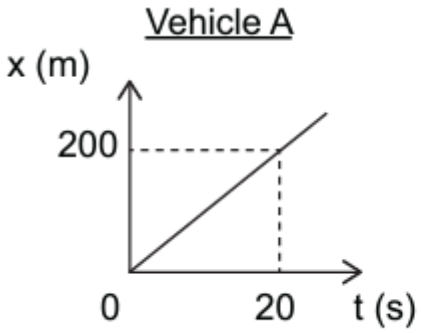
5. A train travels at a constant velocity of 22 $m \cdot s^{-1}$ along a straight railway track towards a level crossing. A mini bus, which is 4,5 m long, is stationary at the level crossing. The mini bus driver decides to cross the track. The mini bus is capable of accelerating at 2 $m \cdot s^{-2}$ and the width of the track is 2,5 m.

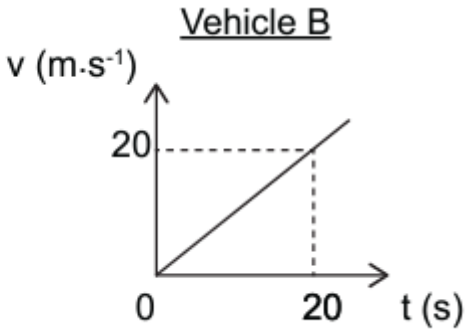


a. Calculate the shortest time it takes the mini bus to cross the railway track. (4)

b. How far away must the train be if the mini bus is to cross the track in front of the train without colliding with it? Assume the train maintains a constant velocity. (3)

6. The two graphs, shown below, are plotted for two different vehicles (Vehicle A and Vehicle B). Study these two graphs and answer the questions which follow.





a. Describe how the motion of Vehicle A differs from that of Vehicle B. (2)

b. How do the displacements of the two vehicles differ after 20 seconds? Which vehicle has travelled further? Justify your answer with reference to the graphs. (3)